

Bounded Communication Fault Tolerance

David Ponarovsky

November 10, 2025

Introduction

Today:

- ▶ Bounded Communication Fault Tolerance
 1. Walkthrough the math
 2. Alternative definition to Expander decomposition
 3. (d, k) - Toric
- ▶ New experimental results.
- ▶ Answering the code teleportation.

Bounded Communication Fault Tolerance

Setting

We allow a non-nosy, strong as we wish, classical computation aside. Yet we would like to limit the communication to a constant number of bits per logical qubit, in each round.

Justification

1. If we don't have a constant overhead in depth, in that model,
 $\Rightarrow$ we also don't have it in the standard model.

Bounded Communication Fault Tolerance

Setting

We allow a non-nosy, strong as we wish, classical computation aside. Yet we would like to limit the communication to a constant number of bits per logical qubit, in each round.

Justification

1. If we don't have a constant overhead in depth, in that model, $\Rightarrow$ we also don't have it in the standard model.
2. In that model, gate teleportation for a constant dimension code is easy. Namely, for the Topological codes the task is reduced to finding a decoding with a constant-size hint.

Max Color

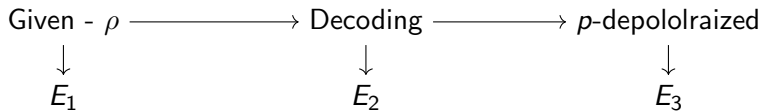
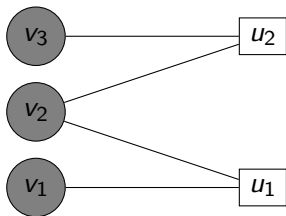
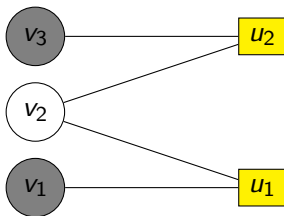


Figure: Illustration of the cycle.

standard model, $t = 0$



simplified (p', q) -model



If the majority of bits in the local view of v_2 are flipped, then after decoding, v_2 will be also (remain) flipped.

What did we learn?

1. Storing time is, likely, not $\Theta(e^n)$, might be $\Theta(L^\alpha)$ for $\alpha \in (0, 1)$.
2. $\Rightarrow$ Implies that the structure of the theorem/key-lemma is different than the known refrigerators.

Future

So What's Next?

1. Moving forward with the experimental work?
 - ▶ More experiments, higher statistical standards?
 - ▶ Accelerate the decoder? Shifting to *c/c++*? Parallelize it with GPUs? Do we have budget?

Future

So What's Next?

1. Moving forward with the experimental work?
 - ▶ More experiments, higher statistical standards?
 - ▶ Accelerate the decoder? Shifting to *c/c++*? Parallelize it with GPUs? Do we have budget?
2. Dividing to colors - Expander Decomposition? What do we know about the expander decomposition of the periodic grid?

Future

So What's Next?

1. Moving forward with the experimental work?
 - ▶ More experiments, higher statistical standards?
 - ▶ Accelerate the decoder? Shifting to *c/c++*? Parallelize it with GPUs? Do we have budget?
2. Dividing to colors - Expander Decomposition? What do we know about the expander decomposition of the periodic grid?
3. How to use the fact our code is a quotient code?